

Can Public Artifacts Substantiate Financial-Agent Alpha? A 98-Work Evidence Audit and Descriptive Spanning Exercise

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Abstract

Can public artifacts substantiate the incremental returns reported for financial agents? A cutoff-bounded screen yields 103 lineages backed by 98 cited works and retains 69 operational formula discovery or trading works, not 69 papers making the same regression-alpha claim. Forty works permit 50 common-task mappings; 29 remain availability-only. A 14-attempt code audit yields no native-agent replication and one monthly adaptation of a released seed, which does not beat the benchmark. We lead with 13 partial mappings anchored in five sources: gross/net median annualized six-factor alpha is 2.44%/0.88%, one is nominally positive at each cost, and none survives within-subset Holm. The other 49 are favorable narrative translations whose failures cannot be charged to their sources. In the full 62-row diagnostic, one M0 proxy survives conditional Holm at the primary five-lag HAC but not at lags 0 or 12; no mapping clears a two-percentage-point familywise materiality test, even gross. Because mappings were frozen after outcomes were inspected and lacked an independent coder, all mapping-based statistics are descriptive conditional diagnostics. The contribution is an auditable evidence ladder showing an artifact gap and limited support from reconstructable components, not a claim that financial agents as a class lack alpha.

CCS Concepts

• **Computing methodologies** → **Multi-agent systems**; • **Applied computing** → **Economics**; • **General and reference** → *Empirical studies*.

Keywords

financial agents, alpha, asset pricing, reproducibility, multiple testing, empirical audit

ACM Reference Format:

Anonymous Author(s). 2026. Can Public Artifacts Substantiate Financial-Agent Alpha? A 98-Work Evidence Audit and Descriptive Spanning Exercise. In *Proceedings of 7th ACM International Conference on AI in Finance (ICAIF '26)*. ACM, New York, NY, USA, 8 pages.

1 Introduction

Financial AI systems now search factor expressions, debate investment theses, retrieve news, and revise portfolios. Their evaluations often report Sharpe ratios, raw trading gains, benchmark outperformance, or alpha [15, 45, 71, 76, 102]. The scientific question is not whether these architectures are interesting. It is whether the evidence released with them supports the claimed incremental return.

That question contains two different failure modes. A system may be impossible to evaluate because its code, data adapter, trained state, or dated output is unavailable. Alternatively, a testable return

path may exist but fail an asset-pricing benchmark. Treating both as zero alpha is wrong; stopping at “not reproducible” is also incomplete. We therefore use the highest-fidelity feasible route and label the estimand at every step. Native or code-backed returns, when available, test a released implementation. When they are unavailable, reconstruction must give the public idea a serious opportunity to work: preserve source-supported content, choose a favorable reasonable implementation, encode ambiguity as variants, and disclose every substitution. A narrative mapping can stress-test an economic story, but its failure cannot be charged to the source.

This distinction narrows the claim. We do not test whether financial agents as a class can discover alpha. We test whether their *public evidence* establishes alpha under an independently specified common task. Our U.S. portfolio results are retrospective historical spanning diagnostics conditional on researcher-authored mappings. They do not demonstrate prospective discovery and cannot correct for prior factor mining or publication selection [23].

Complementary benchmarks use live windows, limit contamination, mask model memory and attribute style, or make factor execution deterministic [8, 94, 105, 109]. We instead contribute a cross-paper 98-work artifact waterfall, a direct-code audit, a source-protective reconstruction contract, and a U.S. asset-pricing ledger. The 13 closest partial mappings provide limited common-task evidence; 49 narrative translations remain exploratory.

2 Evidence Design

2.1 Denominators and inferential targets

The screen covers database inception through 2 August 2026. Its released 22-row route log contains 15 database searches (three query families across arXiv, ACL Anthology, OpenReview, ACM DL, and SSRN), four citation-chasing routes, two deduplication routes, and one record-linked GitHub artifact route. Crossref/OpenAlex resolve duplicates. Literal queries, rules, canonical records, and row rationales are released. Raw hit counts, rankings, and timestamps were not preserved, so this is a cutoff-bounded systematic screen rather than a complete universe.

The registry has 103 lineages, 104 record URLs, and 98 works after six duplicate manifestations are collapsed. All 98 works are cited. Formula discovery (F) and trading agents (T) are retained; benchmarks/audits (B), community software (C), and mechanistic comparators (M) remain visible with reasons. Frozen rules exclude 29 works [11, 65, 66, 87, 97, 103] [9, 13, 14, 34, 36, 57, 89, 95] [20, 29, 33, 38, 52, 55, 59, 62, 84, 88, 92, 106, 108, 110, 112]. The 69 retained works in 67 lineages comprise 30 formula works [4, 10, 15, 28, 31, 45, 50, 54, 64, 68, 71, 74, 76, 79, 91, 100] [3, 22, 32, 43, 46, 49, 67, 72, 78, 80, 93, 96, 101, 102] and 39 trading works [16, 44, 75, 83, 98, 99, 104] [1, 2, 7, 18, 19, 30, 40–42, 47, 48, 51, 56, 61, 69, 70, 73, 81, 82, 85, 86, 107] [5, 6, 12, 17, 25, 27, 37, 39, 63, 77]. Thus 98/69/67 are screened works/retained works/lineages, not replication counts. This retained denominator covers operational methods; Sharpe,

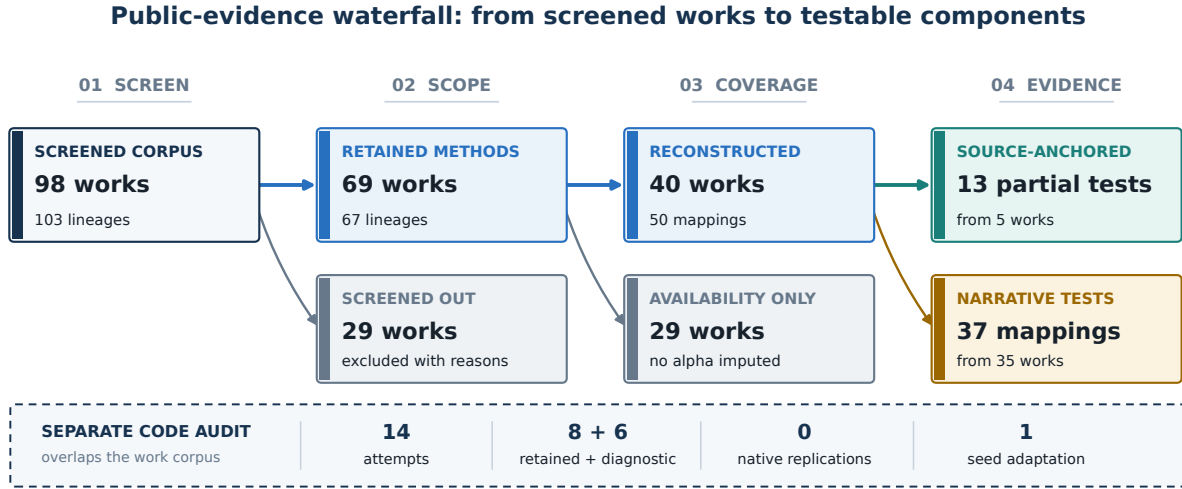


Figure 1: Work-level evidence waterfall. Cards separate the main path from each excluded or availability-only branch: 98 screened works narrow to 69 retained, 40 reconstructed, and five supporting 13 source-anchored partial component tests. The dashed strip is the overlapping code audit. A missing native path is an evidence failure, not a zero return; a mapping tests historical economic content, not agent performance.

raw-return, and benchmark-outperformance claims are not recoded as explicit regression-alpha claims. The common-task regression instead asks whether released or reconstructable content contains incremental return.

We construct 50 mappings for 40 retained works: five works support 13 source-anchored partial component tests and 35 support 37 narrative stress tests. The other 29 remain availability-only because no frozen, supported common-panel mapping exists; no proxy or zero alpha is imputed. Twelve mappings from 11 excluded diagnostic sources produce the prespecified 62-portfolio family. Its legacy discovery list has 55 records and 51 mapped indices (one names two systems). None states our exact monthly U.S. top-1,000 six-factor claim or supports its tested weighting rule, so this is a common-task spanning diagnostic, not 51 claim replications.

The overlapping code route covers eight retained works and six excluded diagnostics (Table 2). We complete zero native-agent replications. One QuantEvolver released seed yields a testable monthly adaptation; 13 blocked attempts never enter a performance denominator. Repository, license, execution, and performance status remain separate fields.

2.2 Artifact and direct-code route

We freeze URLs/revisions and require dated output from the released system for a native path. Of 67 retained lineages, 20 list an artifact, 18 are reachable, nine have a reuse license, and three ship dated native output; none matches the common task. Among 14 targeted implementations, the sole computable adaptation is QuantEvolver’s pinned 60-bar mean-return/volatility seed mapped to 12-month return and 252-day volatility, not a literal expression or trained-agent run. Its annualized six-factor alpha is 0.33% ($t = 0.65$). The other 13 preserve explicit blockers—output, input/adaptor, task,

Table 1: Evidence hierarchy and what each result identifies.

Stage	Evidence/result	Permitted interpretation
Corpus screen	98 works → 69 retained + 29 excluded	All records cited; exclusions remain auditable
Retained coverage	69 works → 40 reconstructed + 29 availability-only	No proxy or zero return imputed to availability-only works
Targeted code audit	8 retained + 6 diagnostic attempts; 0 native; 1 seed adaptation	Adapted-seed result only; no performance claim for 13
Mechanism mapping	50 retained + 12 diagnostic mappings; 13 partial, 49 narrative	Component evidence only; no narrative null charged to a source
Broad diagnostic	133 factors; 0 FWE results	Post-hoc diagnostic, not confirmatory evidence
International forensic audit	40 events; all dominated by one short-held return	Data/implementation alarm; no performance inference

or executability—so the negative performance denominator is one, not fourteen.

2.3 Mechanism mapping and disclosed discretion

When direct evaluation is unavailable, we map public descriptions to point-in-time JKP characteristics [26]. The eight-step good-faith contract prioritizes released expressions, equations, examples/rules, narratives, then summaries; preserves supported signs, lags, interactions, and constraints; gives ambiguity the benefit of the doubt without return tuning; and discloses every substitution. Source-stated direction and long-only or sparse top- k rules are preserved;

Table 2: The 14 targeted implementation attempts. These are the systems whose public code or shipped outputs we actually inspected for a return path; eight belong to the 67-system F/T census.

System	Released object / native task	Audit result
AlphaAgent [71]	China A-share factor-mining code	No shipped dated strategy return series.
AlphaBench [53]	Formula-mining benchmark code	Executable benchmark components, but no dated strategy returns.
AlphaForgeBench [105]	End-to-end strategy-design benchmark	Aggregate metrics only; no dated common-task return stream.
AlphaPROBE [21]	Graph-based factor-mining comparator	No approved JKP adapter or released dated expressions/returns.
ContestTrade [70]	Sequential multi-agent trading code	No replay adapter or shipped common-task returns.
DeepFund [35]	Live fund-investment benchmark	No approved historical reproduction path for the common panel.
FinCon [99]	Sequential multi-agent trading system	Available implementation was unofficial/ambiguous and task-incompatible.
GuruAgents [30]	Nasdaq-100 guru-persona portfolios	Shipped non-JKP backtest is inspectable but incompatible with the common task.
LiveTradeBench [94]	Live stock/prediction-market benchmark	No frozen historical return path compatible with the evaluator.
QuantAgent [85]	Hourly/4-hour HFT across stocks, futures, and crypto	Intraday native task cannot be represented by monthly JKP data.
QuantEvolver [102]	Factor-discovery framework plus seed expressions	Monthly JKP adaptation of the pinned 60-bar return/volatility seed; $\alpha = 0.33\%$, $t = 0.65$. Not the literal expression or a trained-agent run.
R&D-Agent-Quant [91]	General R&D-Agent framework and quant workflow	No shipped U.S. candidate returns or completed JKP adapter.
TradeTrap [90]	Trading-agent reliability harness	Shipped results are a non-JKP diagnostic, not a common-task return path.
Trading-R1 [82]	RL-trained trading-reasoning system	Public repository snapshot is a placeholder without executable system code.

alternatives should be coded before outcomes and remain in the multiplicity family.

A source-protection rule lets released expressions, named rules, and examples speak only to that component. A narrative mapping is a favorable stress test whose failure cannot count as evidence against the source. The public card records supported/supplied elements, omissions, alternatives, outcome/reviewer status, and the negative-evidence boundary; preferred review uses a second outcome-blind coder.

The mappings predate this contract, so we apply only its source-protection rule retrospectively. Each row records the source, rationale, score, portfolio rule, support, omissions, favorable choices, and permitted interpretation. Table 4 labels narrative translations M0 even when favorable.

M1/M2 are reported first as 13 source-anchored partial tests; M0 results belong to our translations. Neither tier supports a verdict on a native agent or original task.

The ledger makes three adverse facts explicit. First, 49 mappings are narrative translations and barred from source-level negative inference; 12 have only partial support, including one released-seed adaptation. A new 13-row review packet supplies page/section or pinned-repository anchors, original tasks and metrics, supported content, and every researcher-supplied change; independent review remains pending. Second, the registry was frozen after U.S. outcomes had been inspected, so its immutable hash does not make coding outcome-blind. Third, no independent second coder was used. The unknown mapping choice set lies outside Holm and bootstrap calculations, which quantify sampling variation only conditional on 62 realized formulas. Every p -value and interval is therefore descriptive, not confirmatory.

We can measure only a lower bound on that discretion. Five sources already have multiple coded mappings, covering 15 candidates. Selecting one mapping per source yields 144 combinations; the same AlphaAgent-inspired quality proxy is the sole Holm discovery in every combination. This does not validate singly mapped formulas, nor does it replace independent coding. It shows only that the headline count is unchanged within the alternatives that happened to be coded.

2.4 Common U.S. task and descriptive estimators

The primary panel is monthly U.S. common equity from August 2001 through December 2024. At each formation month we retain the 1,000 largest positive-market-equity securities, compute signals using information available at formation, and hold for the next month. Most mappings are value-weighted top-minus-bottom deciles; long-only mappings are labeled. Long and short legs each carry unit notional. Candidate and long-only paths use next-month excess returns, so the risk-free return is removed rather than represented by a separate cash sleeve. The locked primary policy assigns zero at the frozen weight when a held next-month return is missing; this avoids selecting or renormalizing holdings using future coverage. We report missing gross-weight exposure and a deliberately severe stress assigning -100% to a missing long return and $+100\%$ to a missing short return. It is not an expected delisting-return estimate.

Turnover uses drifted pre-trade weights. The locked 10 bp one-way charge is a transparent reference point, not a security-specific estimate; we therefore report the entire 0/5/10/25/50-bp curve and candidate break-even costs. Borrow, financing, locate constraints, nonlinear impact, and capacity are omitted. Benchmark factors are gross, so costed-candidate alpha is a deliberately asymmetric, conservative comparison rather than two equally costed tradable products.

The primary benchmark contains the market plus JKP analogues of size, value, investment, profitability, and momentum [26]. For candidate i we estimate

$$r_{i,t}^{\text{net}} = \alpha_i + \beta_i' f_t + \varepsilon_{i,t}.$$

The primary Newey–West lag is the automatic rule $\lfloor 4(T/100)^{2/9} \rfloor$ (five at $T = 281$) [60]. We report two-sided nominal tests, Holm correction across all 62 candidates [24], and paired moving-block maximum- $|t|$ inference with 2,000 bootstrap replications and a six-month block. Block lengths three and twelve and HAC lags 0, 3, 6, and 12 are sensitivities. A simultaneous lower bound of at least two annual percentage points is a pragmatic reporting threshold selected by the project owner after U.S. outcomes were available; it is not a preregistered decision rule.

Table 3: The five papers underlying the 13 source-anchored partial component tests. The anchor applies only to the listed content, not the full agent, score, or original evaluation.

Source (N mappings)	Original idea and task	Component implemented in our common task
EFS [54] (3)	LLM evolutionary factor search for sparse top- m portfolios on Fama–French and US50/HSI45/CSI300 datasets.	Momentum/low-volatility/breakout score; reversal/low-noise score; and a sparse equal-weight top-five variant.
QuantEvolver [102] (1)	Reinforcement fine-tuning generates executable factor-DSL expressions across market benchmarks.	Monthly JKP adaptation $\text{ret}_{12,1}/ \text{rvol}_{252} $ of the released 60-bar return/volatility seed; horizons and portfolio formation are ours.
Alpha-Jungle [68] (2)	LLM-guided MCTS searches symbolic stock factors using price, volume, and volatility operators.	A price/volume/momentum score and a trend-after-volatility-compression score based on documented examples.
FAMA [45] (1)	Neural-symbolic factor mining evaluated by RankIC/RankCIR and an S&P 500 investment simulation.	A simple interpretable value–momentum–profitability score motivated by the paper’s factor examples.
GuruAgents [30] (6)	Prompt-guided agents emulate five named investor philosophies in a Nasdaq-100 backtest.	Graham, Buffett, Greenblatt, Piotroski, and Altman characteristic screens plus their equal-weight ensemble.

Table 4: Mapping fidelity and U.S. results by tier.

Evidence tier	N	Median alpha	Cond. Holm +
M0 narrative translation	49	−0.10%	1
M1 example/motif support	6	−0.71%	0
M1 named-rule support	6	3.12%	0
M2 released-seed adaptation	1	−0.28%	0

Annualized six-factor alpha net of 10 bp one-way costs. Holm values are conditional on the realized mappings and are descriptive.

3 Results

3.1 Descriptive U.S. historical spanning exercise

The evidentiary lead is the 13-row partial subset: at 10 bp, median alpha is 0.88%, seven estimates are positive, one is nominally positive, and none survives within-subset Holm. The released-seed adaptation has alpha −0.28% in this costed portfolio implementation; the separate direct-code series is gross, uses a different portfolio return path, and has alpha 0.33% ($t = 0.65$). Thus neither route supplies adjusted evidence of incremental return.

The full 62-row diagnostic is broader but less attributable. All paths are executable for 281 months. At 10 bp, median annualized alpha is −0.05% (IQR [−1.26%, 2.05%]); 30 estimates are positive and 6 are nominally positive. One M0 mapping has conditional Holm, maximum- $|t|$, and White-style family $p = 0.0105$ below .05. These agree only conditional on the realized mappings; they do not account for mapping discretion. No simultaneous lower bound exceeds two percentage points. The largest estimate, 6.93%, is a *researcher-authored AlphaAgent-inspired quality proxy*; the software did not generate it, and its simultaneous lower bound is 0.48%.

The conditional result is fragile to serial-correlation choices. Across fixed Newey–West lags 0/3/6/12, nominal positive counts are 5/6/6/6 and Holm counts are 0/1/1/0: the adjusted count is one only at lags 3 and 6. Block-length changes leave the maximum-statistic count at one, but cannot cure mapping or publication-selection uncertainty.

Costs change signs but not the central material-alpha finding. Median alpha moves from 2.15% gross to 1.01% at 5 bp and −0.05% at 10 bp; positive counts across 0/5/10/25/50 bp are 46/42/30/18/10, while conditional Holm counts are 1/1/1/0/0. Among 46 gross-positive mappings, median break-even cost is 21.4 bp (IQR 8.2–42.1). No cost level, including zero, has a Holm result for $H_0 : \alpha \leq 2\%$; hence the familywise material-alpha conclusion does not rely on

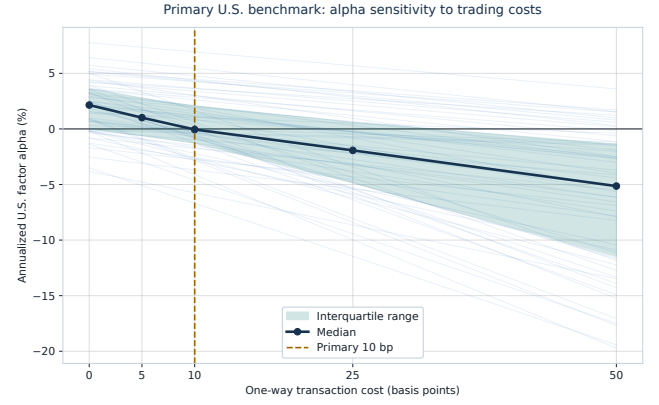


Figure 2: Annualized six-factor alpha across one-way trading-cost assumptions. Thin lines are candidate mappings; the band is the interquartile range. Benchmark factors remain gross.

the 10 bp calibration. Security-level impact and capacity remain unidentified.

Missing-return exposure is small in weight but economically nontrivial under an extreme imputation. The median mapping’s mean missing gross-weight share is 0.20% (maximum across mappings 0.43%). The position-adverse unit-move stress changes family median alpha to −4.84%, leaves 4 positive estimates, and leaves zero nominal or conditional Holm positives. This does not identify delisting returns; it shows that zero fill is consequential and that adverse missing outcomes do not create evidence of alpha.

Table 5 anchors the gross result. Median $\hat{\alpha}$ is 2.15%; 46 estimates are positive, seven nominal, and one Holm-positive for $H_0 : \alpha = 0$. Of 33 point estimates above 2%, three reject $H_0 : \alpha \leq 2\%$ unadjusted and none survives Holm. The M0 survivor has gross/net Sharpe 0.80/0.74 and $\hat{\alpha}$ 7.76%/6.93%; it is not agent output. The largest partial component has 5.15% gross and 4.46% net alpha but no Holm result. The direct QuantEvolver adaptation has gross Sharpe 0.316 and $\hat{\alpha} = 0.33\%$ ($t = 0.65$). These realized returns are not an independent out-of-sample discovery test because mapping lacked a clean holdout.

These results support two deliberately separated findings. The 13 source-anchored partial mappings provide no familywise evidence that their documented content adds U.S. alpha under the common task. The other 49 show what happens to generous characteristic

Table 5: Gross-to-net alpha thresholds and numerical anchors for Figure 2.

Panel A: familywide threshold audit								
Cost	Med. $\hat{\alpha}$	Med. SR	$\alpha > 0$			$\alpha > 2\%$		
			Point	Raw	Holm	Point	Raw	Holm
0 bp	2.15%	0.214	46	7	1	33	3	0
5 bp	1.01%	0.153	42	7	1	21	1	0
10 bp	-0.05%	0.123	30	6	1	16	1	0
25 bp	-1.93%	0.022	18	1	0	10	1	0
50 bp	-5.14%	-0.136	10	0	0	1	0	0

Panel B: selected gross and 10-bp mapping rows						
Evidence line	Gross (0 bp)			Net (10 bp)		
	SR	$\hat{\alpha}$	t	SR	$\hat{\alpha}$	t
Largest $\hat{\alpha}$: P-D733C6 (M0)	0.800	7.76%	3.86	0.739	6.93%	3.45
Largest grounded: P-54358A (M1)	0.661	5.15%	2.38	0.609	4.46%	2.07
Released-seed mapping: P-93491E (M2)	0.308	1.63%	0.92	0.210	-0.28%	-0.16
Direct seed adaptation (gross only)	0.316	0.33%	0.65	—	—	—

U.S. mapping rows cover 2001–2024. In the $\alpha > 0$ columns, Raw means positive $\hat{\alpha}$ with two-sided HAC $p \leq .05$ and Holm adjusts across 62 mappings. For $\alpha > 2\%$, Raw and Holm use one-sided HAC tests of $H_0 : \alpha \leq 2\%$; that materiality threshold was selected after outcomes and remains descriptive. SR annualizes next-month realized returns, which are one-period-ahead but not an independent discovery holdout. The direct released-seed adaptation is a separate gross 1999–2024 path. Panel B rows are selected by evidence role, not significance.

translations, but cannot reject the originating papers. Neither result shows what an unavailable native agent would earn or establishes prospective discovery.

3.2 Post-hoc broad-factor diagnostic

The post-hoc diagnostic uses the market plus 132 JKP factors. The source panel ends in December 2021; after shifting factors to return months and a 120-month trailing window, evaluation spans August 2011–January 2022 (126 months), not 2024.

Each month selects the ridge penalty on 96 past fit and 24 validation months, refits on all 120, and emits a one-month-ahead residual. The 5,000-replication block bootstrap resamples already-estimated residuals; it does *not* rerun rolling estimation or penalty selection. Conditional on this possibly optimistic realized path and designed after primary outcomes, it is descriptive, not robustness confirmation.

There are 23 positive estimates and one nominal result, but zero conditional Holm or maximum- $|t|$ results and zero simultaneous lower bounds above two percentage points. The best raw p -value is 0.0025; its Holm value is 0.152 and maximum-statistic value is 0.091. This is directionally consistent with spanning, but remains a post-hoc diagnostic over a truncated period.

3.3 International extension: forensic result, not alpha evidence

The international extension is excluded from headline performance inference. Thirty-five paths cross nonpositive NAV, producing 40 events (31 France, five Canada, two Germany, one Britain, one Italy). Security-level reconstruction matches every event; in all 40, one extreme positive return held short causes at least half the loss. Two France month-cells contain 24 events.

Return caps of +100%/+200%/+500%/+1,000% leave 0/1/1/7 events, but are not admissible corrections. Corporate actions, return construction, stale prices, or short-weight interactions may be responsible. Without independent security-level validation, this is a data/implementation alarm; the ledger retains every event, identifier, weight, return, contribution, error, and cap diagnostic.

4 What the Evidence Does and Does Not Establish

Scientific value lies in an auditable 1-of-14 code-backed performance path; a fairness rule giving supported components favorable tests while restricting attribution; and cost, missing-return, factor, and multiplicity diagnostics exposing the fragility of raw-return or nominal-test conclusions.

The design has firm limits. Many systems postdate the December 2024 return panel, and mappings were frozen in 2026 after outcome inspection. The exercise cannot distinguish discovery from repackaging of known characteristics. Holm adjustment covers sampling multiplicity only across the 62 realized formulas; it does not cover researcher mapping choices, the literature’s search process, unpublished mappings, or publication selection. Without an independently coded, outcome-blind reconstruction, performance results remain descriptive.

Common-task comparability trades away language, memory, retrieval, debate, native timing/execution, dynamic allocation, non-equities, and tool adaptation. Mapping failure cannot prove native-agent failure; success cannot be attributed to an agent. The strongest-looking row is M0, precisely where attribution is forbidden. The page-anchored packet makes the 13 partial mappings adjudicable, but only a prospective outcome-blind reimplement can remove this limitation.

Benchmark practice outside finance offers a model: ABC separates task/outcome validity and reporting [111]; Terminal-Bench uses containerized coding tasks, executable tests, human solutions, and failure analysis [58]. Financial-agent releases additionally need pinned code/environments; dated signals, positions, or returns; universe, timing, cost, financing, model, prompt, and seed definitions; search breadth and failures; and provenance or ablation separating agent contribution from factor libraries and human search spaces.

5 Reproducibility and Audit Trail

The ledger contains the 22-route search protocol; 103 lineages, 104 URLs, and 98 dispositions; 14 code attempts; the five-paper/13-row source-anchor packet; 62 formulas/decisions; aggregate U.S. returns; cost, missing-return, broad, and international diagnostics; scripts, tests, and hashes. It states that raw hits were not preserved, mappings were not outcome-blind, independent review is pending, and restricted security data are referenced by path and hash.

Deterministic builders regenerate macros/figures from verified outputs. Tests cover timing, weights, drift, costs, inference, mapping sensitivity, citations, and event reconstruction. Claims do not depend on an unsubmitted appendix.

Responsible use and AI assistance. No orders were transmitted. Results are retrospective research evidence, not investment advice. Generative AI assisted literature organization, code drafting, diagnostic review, and prose editing; the human author is responsible

for source selection, mappings, protocol choices, statistical specifications, citation verification, and all reported claims.

6 Conclusion

Current public artifacts do not establish broad, economically material U.S. financial-agent alpha. The 98-work screen narrows to 69 retained, 40 reconstructed, and five supporting partial component attribution. The code route yields zero native-agent replications, one testable seed adaptation, and no benchmark beater; 13 blockers are evidence failures, not negative returns. Of 13 source-anchored partial mappings, one is nominal and none conditionally Holm-positive; 49 narrative tests cannot count against their sources. The sole full-family conditional result is HAC-fragile, no alpha clears the familywise materiality threshold even gross of costs, the adverse missing-return stress has no positive test, and the post-hoc broad diagnostic has no adjusted result. International performance is excluded after a plausibility failure.

The paper's contribution is therefore constructive rather than categorical. It supplies an evidence hierarchy and an inspectable good-faith workflow for giving unreleased ideas their strongest reasonable test, while stopping interpretation exactly where fidelity and design no longer permit attribution.

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